

CATALIN MIRISAN

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EDUCATION

B.Sc. in Informatics (Romanian Program)

Babeş-Bolyai University, Cluj-Napoca | Sep 2024 – Jun 2027

Bachelor's Thesis: "Synchronized and Secure Storage System with Data Deduplication"

Relevant Coursework: Cryptography, Operating Systems, Networks, Databases, Algorithms, Security, Distributed Systems

PROFESSIONAL EXPERIENCE

Artificial Intelligence Intern

Globant | Cluj, Romania (Hybrid) | Nov 2025 - Jan 2026

- Worked on Artificial Intelligence projects and gained hands-on experience with Python-based AI development
- Technologies: Python, AI, Machine Learning, Software Development

Intern - Environment Packaging Data Management

Tenaris Silcotub | Zalău, Sălaj, Romania (On-site) | Jul - Sep 2026

- Digitalized and centralized all packaging data placed on market by Silcotub for environmental reporting
- Developed unified digital platform to collect, validate, store, trace, and report packaging data efficiently
- Collaborated with Environment Department to improve data accuracy and compliance with regulatory requirements
- Key Skills: Data Management, Environmental Compliance, Data Collection, Process Digitalization

PROJECTS & TECHNICAL WORK

Homelab Media & Security Stack

Personal Project | Aug 2026 - Present

- Self-hosted media stack on Proxmox with 8 unprivileged LXC containers: qBittorrent, Radarr, Sonarr, Seerr, Jellyfin, Prowlarr, ClamAV, Uptime Kuma
- Implemented Zero-Trust remote access via Tailscale with VPN kill switch (Mullvad/WireGuard) enforced via iptables + routing rules
- Automated malware scanning (ClamAV) triggered on every download; IPv6 blocking, firewall rules, automated backups
- Technologies: Proxmox, Linux, Bash, Docker, WireGuard, Tailscale, iptables, Security Automation
- *Repository:* github.com/MirisanCatalin/homelab-media-security-stack

Secure Deduplicated Sync (Bachelor's Thesis)

Babeş-Bolyai University | Aug 2026 - Jun 2027

- Designing synchronized storage system with data deduplication combining applied cryptography and distributed sync
- Focus: Convergent encryption, content-defined chunking, HMAC integrity verification, threat modeling
- Status: In progress - repository to be public after thesis defense

TECHNICAL SKILLS

Languages: Romanian (Native), English (C1 Advanced)

Programming: Python, Bash, Go, Rust, Java, C, C++

Systems & Virtualization: Linux (Debian, Ubuntu), Proxmox VE, LXC, Docker, systemd, Virtualization

Security & Cryptography: WireGuard, Tailscale, iptables, ClamAV, Applied Cryptography, Threat Modeling, Penetration Testing Basics

Tools & Platforms: Git, GitHub, Ansible, Bitwarden, Wireshark, Nmap, Proxmox, Docker

Databases: SQL, PostgreSQL, MySQL, NoSQL basics

SOFT SKILLS & ATTRIBUTES

- Technical Documentation - Clear written communication of complex systems and processes
- Systematic Debugging - Methodical problem-solving and root cause analysis
- Security Mindset - Privacy-first thinking, zero-trust principles, compliance awareness
- Teamwork & Collaboration - Effective communication in hybrid work environments

INTERESTS & ACHIEVEMENTS

- Deep interest in cybersecurity, cryptography, and secure system design
- Active learner: homelab experimentation, open-source contributions, continuous skill development